

Configurable RS232 To TTL To I2C ADAPTOR

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RS232 signals cover a much longer distance than standard TTL and I2C signals. Also, RS232 signals have better noise immunity. RS232 is a well-known standard supported by many popular integrated circuits (ICs) and therefore is still widely used. Sometimes we want to send TTL and I2C signals over a longer distance. This article describes how these signals can be extended through RS232 lines.

Circuit and working

Fig. 1 shows the circuit of configurable RS232 to TTL to I2C adaptor. RS232 signals are available on the standard 9-pin connector, also known as DB9 connector (CON1). There is a wide variety of configurations available for RS232 signals on different peripherals. So we have provided CON2 and a 4-way or 8-pin DIP switch (DIP1) for various

configurations. CON2 is a single-line connector from which RS232 signals can be taken for interfacing external circuits.

MAX232 (IC1) is a dual driver/receiver that includes a capacitive voltage generator to supply TIA/EIA-232-F voltage levels from a single 5V supply. It has two receivers and two transmitters compatible with RS232 signals.

TTL equivalent signals from the interface are available at CON3. These signals drive LED1 through LED4 through gates N1, N4, N5 and N6 of hex inverter 7405 (IC2) for visual indications. The 7405 should be with low input current and capable of driving the LEDs.

Switches S1 and S2 are used to manually check the signals on LED1 and LED2 when DIP2 pins 1 and 8 and pins 3 and 6 are shorted. DIP2 is used to provide different configura-

PARTS LIST

Semiconductors:

- IC1 - MAX232 driver
- IC2 - 7405 hex inverter
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4-R6 - 560-ohm
- R2-R3 - 4.7-kilo-ohm
- R7, R8 - 10-kilo-ohm

Capacitors:

- C1-C4 - 1 μ F, 16V electrolytic
- C5 - 100 μ F, 16V electrolytic
- C6, C7 - 0.33 μ F ceramic disk

Miscellaneous:

- CON1 - DB9 COM port connector
- CON2, CON3 - 6-pin berg strip connector
- CON4 - 2-pin connector terminal
- CON5 - 4-pin berg strip connector
- S1, S2 - Toggle on/off switch
- DIP1, DIP2 - 4-way DIP switch
- 5V DC regulated power supply

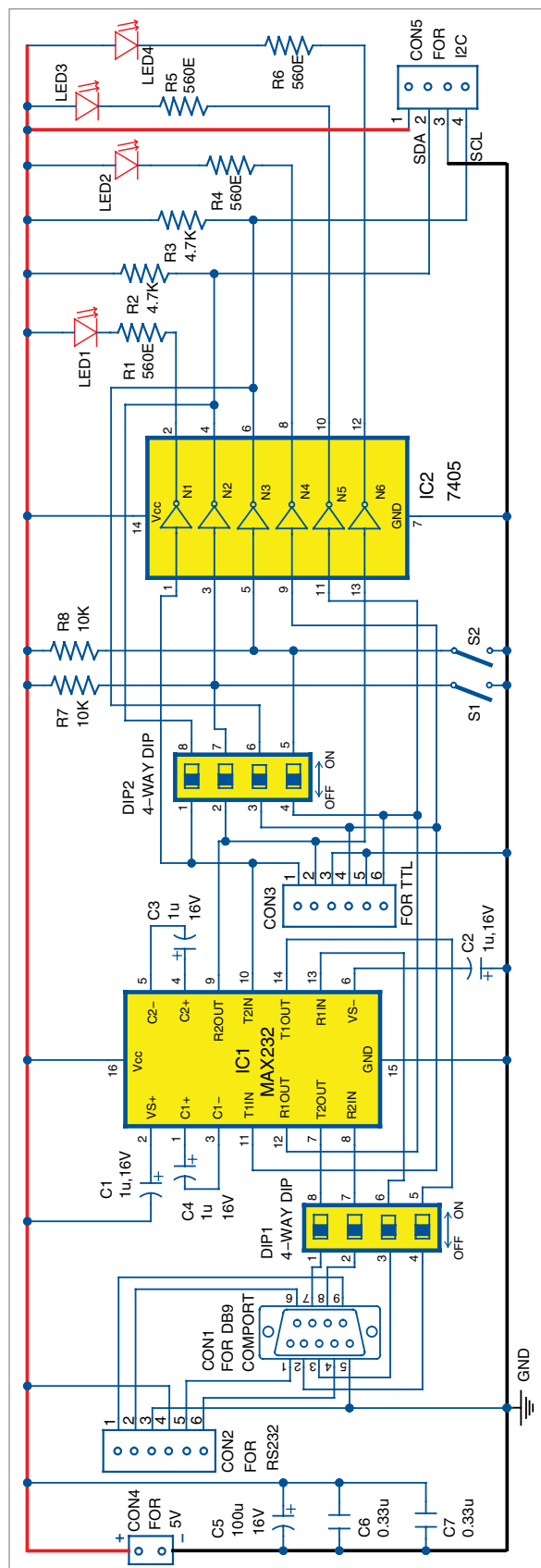


Fig. 1: Circuit of configurable RS232 to TTL to I2C adaptor